$\frac{}{\sqrt{2}} \quad \frac{}{\sqrt{2}}$

10 **B**

$$\frac{GM}{x^2} = \frac{G(4M)}{(d-x)^2}$$

$$\frac{(d-x)^2}{x^2} = 4$$

$$\frac{(d-x)}{x} = 2$$

$$x = \frac{d}{3}$$

11 **C** work done by external force $= m \times \Delta\phi$